

Mansalvic Consulting LLC

System Specification & QA Testing Master Guide

Document Version: 2.0 | **Status:** Production Ready & Verified | **Date:** September 2026
Target Audience: QA Engineers, Manual & Automation Testers, Technical Leadership
Core Scope: Web Application Architecture, Dynamic Consultation Workflows, Google Meet / Zoom Appointment Scheduling, Firestore Real-time Persistence, and Admin Telemetry.

1. Executive Product Overview & Capabilities

Mansalvic Consulting LLC is an enterprise technology consultancy platform connecting North American businesses with high-velocity engineering solutions. The web application delivers three primary lines of service: IT Staff Augmentation & Dedicated Engineering Teams, Custom Full-Stack Application Development, and 24/7 Application Maintenance with guaranteed SLAs. Built upon an eco-sustainable visual theme, the platform combines modern web engineering with an interactive business assessment questionnaire and a fast-track video appointment booking calendar operating on Google Meet and Zoom.

2. Core Business Requirements & Constraints

ID	Requirement Title	Business Rule & Acceptance Constraint
BR-01	Meeting Format	All consultation calls are conducted exclusively via Video Meeting (Google Meet or Zoom). Regular telephone calls are completely eliminated.
BR-02	Operating Schedule	Appointments are restricted strictly to Monday through Friday, between 9:00 AM and 5:00 PM EST, generated in 30-minute intervals.
BR-03	Calendar Availability	Weekends (Saturday/Sunday) and past dates are completely disabled. Business days with open hours are clearly marked with green indicator dots.
BR-04	Concurrency & Collision	Real-time checking against the Firestore 'appointments' collection ensures already confirmed slots are greyed out and unselectable.
BR-05	Fast-Track Skip Flow	Every consultation button on every page (About, Hero, Services, Drawers, Header, Footer) must provide an option to skip questions straight to the meeting calendar.
BR-06	Requirements Pre-fill	Any written requirements typed into free-form notes before clicking skip must automatically populate the appointment booking form.
BR-07	Zero Blank Screen Policy	Defensive parameter type-checking and React Error Boundaries ensure no synthetic event or runtime error crashes the UI to a white screen.
BR-08	Data Persistence	Confirmed bookings are written simultaneously to Firestore 'appointments' and 'leads' collections with complete attendee details.

3. System Architecture & Testing Environment

The frontend is developed in React 18 with Vite 5 and Lucide React icons. Data operations interface directly with Google Firebase Firestore. The application can be verified in both developer mode and production emulator mode:

Component	Local Address / URL	Operational Purpose
Vite Dev Frontend	http://localhost:3000	Hot-reloading development server for UI testing
Firebase Emulator	http://localhost:5000	Static production build served from public/ directory
Homepage & Drawers	http://localhost:3000/#home	Primary landing page, metrics, animated scene, and leaf drawers
Services Directory	http://localhost:3000/#services	Full service catalog with technical capability breakdowns
About Us Page	http://localhost:3000/#about	Corporate background, Ohio headquarters, global delivery deck

Admin Telemetry	http://localhost:3000/#admin	Passcode-protected portal monitoring clicks, reads, and leads
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4. Detailed Functional Specifications

4.1 Header Navigation Bar (Header.jsx)

The fixed header includes the brand logo, navigation links (Home, Services with direct dropdown anchor links, About Us), and three distinct action buttons: 'Schedule Meeting' (opens the video calendar directly), 'Request Talent' (launches modal pre-selected to Staffing), and 'Start Consultation' (launches modal at Step 1).

4.2 Hero Section & Ecological Landscape (Hero.jsx & EcoLandscapeScene.jsx)

Features high-impact enterprise metrics (500+ staffing network, 99.98% SLA uptime, 40% compute optimization) and a custom animated SVG scene. Dual consultation entry points allow visitors to either 'Start Interactive Business Assessment' or click 'Schedule Video Meeting' to skip directly to booking.

4.3 Interactive Leaf Services Accordion (Services.jsx)

Presents three interactive cards with animated Duolingo-style mascots: Leo (Staffing), Alex (Development), and Pip (Maintenance). Clicking any card expands only that specific drawer. Inside the drawer, users can calibrate parameters with selection pills, enter specific project requirements in a free-form notes textarea, click 'Schedule Video Meeting' to fast-track directly to booking with their notes pre-filled, or click 'Launch Full Consultation Dialogue'.

4.4 Full Services Directory Page (ServicesPage.jsx - #services)

A comprehensive directory detailing capabilities, deliverables, SLA tiers, and pricing models. Each card contains both 'Request Proposal' and a direct 'Schedule Meeting' video button. The bottom callout banner provides dual actions: 'Launch Consultation Dialogue' and 'Schedule Video Meeting (Skip Questions)'.

4.5 About Us Page (AboutPage.jsx - #about)

Highlights Mansalvic's US presence in Columbus, Ohio, and international engineering network. The 'Sustainable IT Philosophy' card provides both deck request and 'Schedule Video Meeting (Skip Questions)' buttons. The bottom banner features dual action buttons for both the consultation dialogue and direct meeting scheduling.

4.6 Dynamic Consultation Questionnaire Modal (QuestionnaireModal.jsx)

A 6-step guided consultation flow. At the very top of every step, a permanent shortcut banner allows users to bypass all questions at any time: 'Prefer to talk directly? Skip questions and schedule a 1-on-1 meeting on Google Meet or Zoom'. In Step 1, prospects choose their service and can input custom notes. In Steps 2 to 5, tailored questions appear based on the active service, each with a notes field and an active 'Skip Questions & Schedule Meeting' button. Step 6 captures contact details and saves the lead to Firestore with an instant consultation receipt.

4.7 Appointment Booking Calendar (AppointmentBookingCalendar.jsx)

A dedicated calendar scheduling interface. Prospects select their preferred platform (Google Meet or Zoom), pick an available weekday marked with a green indicator dot, and select an open 30-minute time slot between 9:00 AM and 5:00 PM EST. Booked slots are queried in real time from Firestore and greyed out. Upon submitting contact details, records are created in Firestore 'appointments' and 'leads' collections, and an official booking confirmation ticket is displayed.

5. QA Test Cases & Acceptance Criteria Matrix

Case ID	Objective	Execution Steps	Expected Result & Pass Criteria
TC-NAV-01	Header Direct Meeting	Click 'Schedule Meeting' in header.	Modal opens directly in calendar view. Zero questions displayed.
TC-NAV-02	Header Consultation	Click 'Start Consultation' in header.	Modal opens at Step 1 allowing service selection.
TC-ABT-01	About Banner Skip	Navigate to /#about, scroll to banner, click 'Schedule Video Meeting (Skip Questions)'.	Calendar modal opens directly. Google Meet / Zoom selector visible.
TC-ABT-02	About Philosophy Skip	On /#about, click 'Schedule Video Meeting' in Sustainable IT card.	Calendar modal opens directly with general service pre-selected.
TC-HERO-01	Hero Direct Meeting	Click 'Schedule Video Meeting' on the homepage hero.	Calendar modal opens directly without questionnaire.
TC-SVC-01	Services Card Meeting	On /#services, click 'Schedule Meeting' on Staff Augmentation card.	Calendar modal opens pre-configured for Staffing service.
TC-LEAF-01	Leaf Single Expansion	Click Staff Augmentation leaf card.	Only Staffing drawer expands. Sibling cards remain collapsed.
TC-LEAF-02	Leaf Note Pre-Fill	Type 'Need 2 Senior DevOps engineers' into drawer notes, then click 'Schedule Video Meeting'.	Calendar modal opens. The typed note is visible and preserved in the requirements review box.
TC-MOD-01	Modal Top Shortcut	Open questionnaire. On Step 1, click 'Schedule Video Meeting' in top green banner.	View instantly switches to the appointment calendar.
TC-MOD-02	Deep Question Skip	Proceed to Step 3, enter custom text, click 'Skip Questions & Schedule Meeting'.	View switches to calendar. Custom notes are retained.
TC-MOD-03	Step 1 Validation	On Step 1, click 'Next Step' without selecting a service.	Validation error: 'Please select one of the services above to proceed.'
TC-MOD-04	Questionnaire Submit	Complete all 6 steps with valid contact info and submit.	Lead saved to Firestore. Receipt displays unique Reference ID.
TC-APT-01	Weekend Disabling	Open calendar. Attempt to click Saturday or Sunday dates.	Weekend cells are greyed out, have cursor: not-allowed, unclickable.
TC-APT-02	Business Day Marker	Inspect Monday-Friday cells on current month.	All available business days display green indicator dots.
TC-APT-03	30-Min Slot Generation	Select an available weekday.	16 slots appear (09:00 AM to 04:30 PM EST).
TC-APT-04	Platform Toggle	Click 'Zoom', then click 'Google Meet'.	Platform toggles cleanly with active pill styling and helper text updates.
TC-APT-05	Email Validation	Select date and time, leave email blank, attempt submit.	Submit is prevented with validation alert for required email.
TC-APT-06	End-to-End Booking	Select platform, date, time, enter Full Name and Email, submit.	Booking ticket confirms ID, format, date, and EST time. Firestore record created.
TC-APT-07	Slot Collision Guard	Re-open calendar for date booked in TC-APT-06.	The booked time slot is greyed out (.unavailable) and disabled.

6. Tester Sign-Off Checklist

Before certifying a release candidate, QA testers must verify:

- **Zero Runtime Crashes:** No unhandled React synthetic event exceptions or white screens.
- **Full Skip Availability:** All consultation buttons across all pages have an immediate path to schedule a video meeting.
- **Video Exclusivity:** Zero occurrences of 'phone call' terminology in the appointment booking workflow.
- **Platform Storage:** Both Google Meet and Zoom selections correctly persist in Firestore records.
- **Collision Locking:** Booked appointments correctly block the selected time slot on subsequent queries.